



From a young age, Department of Agriculture, Fisheries and Forestry (DAFF) Senior Entomologist Dr Mike Elias loved insects. He was the kid who would collect bugs and bring them home, and he never grew out of it. Today, that lifelong fascination supports the scientific work that helps protect Australia from exotic pests and diseases.

“I still like collecting insects; I find them fascinating,” said Mike, who now works at the new state-of-the-art Australian National Insect Collection (ANIC) facility based at the Commonwealth Scientific and Industrial Research Organisation (CSIRO) in Canberra.

There, he is surrounded by about 12 million insect specimens that help scientists at DAFF’s eight laboratories identify pests and stop their entry into Australia. The collection (largely owned by CSIRO) is an important part of Australia’s biosecurity system, helping scientists quickly and accurately identify insects that could pose a risk to our industries, environment and communities.

Mike’s primary role is to help DAFF scientists identify insects at the species level, by sharing specimens and providing advice. He also leans on the expertise of CSIRO entomologists and taxonomists.

While never intending to make insect identification his career, he has been working in diagnostic roles since completing his doctorate in new plant bugs.

Working for DAFF in Sydney and Canberra, he later moved to a state-based insect collection facility in Orange before moving to ANIC. Throughout his career, Mike has helped build Australia's insect identification capability and scientific knowledge.

"We have one of the best biosecurity systems in the world, and what we do as DAFF scientists behind-the-scenes is of critical national importance," he said.

"Being a functional entomologist is very complex work; at DAFF, you need about two years of training to be good at it.

"What I love is the variety. One day it might be species identification, another day it might be research or providing policy advice."

Another plus for Mike is he has an insect named after him. It's called *Panaetius eliasi* – a rare species of Australian shield bug he discovered in New South Wales.

"I found it in a swamp in Smiths Lake and gave it to my colleague, who named it after me," he said.

For Mike, discoveries like these are a reminder that safeguarding Australia's unique species starts with a strong biosecurity system.

Behind every identification is a network of scientists, researchers and biosecurity professionals working to strengthen our biosecurity system, because Australia is worth protecting.

Learn how DAFF is helping build a stronger, more resilient system for the future: [Biosecurity 2030 roadmap](#).

Source: <https://www.agriculture.gov.au/about/news/meet-the-scientist>